John H Clippinger “Workshop part 1: Introduction to Applying Active Inference“

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okay welcome back everyone we are here in part one of the workshop that John clippinger and our partners at first principal first are coordinating so with that for the next session please John to you and take it away well thank you very much um I would like to just to share a a deck with the the group here if I could um and uh and provide a sort of an overview of of um what we'll be talking about in this particular session can you see that now is this is this good for you okay great um and we'll be anticipating with Andrew is uh sure will be coming on and providing the tutorial but what I would like to do is in in the first inst of the is is being able to provide an overview of why active inference uh is important why active inference agents this whole perspective that we're taking here is worth people's attention and pay uh and being concerned about um and what it has to offer to address some of the the the the limitations of the current uh AI systems for example large language models and some of the uh agenda architectures have come out of them um and and in that in that context what I looking at is said what is the current state of large language models and and agent uh an agents and I think you there's a lot of activity that's been talked about but there there's still no Mission critical applications yet there's a lot of an exploration you have people like um Salesforce developing Little Einstein that can do uh various kinds of uh workflow automation you have co-pilot and Microsoft but it's not it's not it's not really core critical not to say that it won't be but it it is still uh is still on the experimental phase it's still a back box we do not know real lot how it works hence you have issues of data Pro Providence you have a question of how to govern it um and there's more recently there's a concern about you've invested a trillion dollar but there's an unknown business model coming out so there's a lot of money that's going into it but it's not clear that the current framing of uh of uh AI um as it's now known is really is going to be viable and part of the issue here is is that we and I love this example from The New Yorker is like who's modeling who we're actually putting data into the models we're trying to write representations and we have that interaction but there's not necessarily an abstraction that's coming out so it's this is kind of this uh relationship of of seeing ourselves mirroring ourselves and going and this sort of cycle um and what we have now is a certain hype cycle so it's around the peak of the this is 2023 from the uh Gartner group and it's the generative AI is at the top of the hype hype cycle and then if it's hard to read this particular uh graph but you'll see that principle AI is is U basically um on the early cycle it's on the early phase of it but it's not seen as taking very long where some like uh you're going to have AGI is seen as being like 20 years out it's the top of the hyp cycle I think if we update this there's now I think it's starting to be appreciated that the generative models and the expectations and the kind of problems and the cost associated with them and the lack of improvement the the the lack of degree of improvement through successive models is starting to show through so the the Next Generation open AI shift strategy is a rate of GPI improveing slow in other words the new Orion model is not did not have the level of improvements that they were expecting I think that is something to take an account um and hence the cost of of of training and inferencing is extremely high and and with that the the M the fact that you have this enormous consumption of energy uh that Microsoft is going to have to recommission Three Mile Island and drive one of their own data centers this shows that the Imp practicality of having to um have so much computation um in order to have work and make these models work and I I think that this is starting to be recognized within the the the limitations of large language models and sort of the the the parametric approach to to uh modeling uh AI modeling and I think just to start to step into what what we have now what we're looking at I I love this is another a cartoon from the the New Yorker and since says to think it all began with letting autocomplete finishing our sentences and well autocomplete is a sort of for Markoff chain so we are we are having the sort of chains on top of chains that we've been able to to develop some pretty rich representations of of of human uh activity text and actually forms of cognition but there's not an underlying model there and so what we're looking at there may be some foundational issues that are coming up in in very large language models and that the fact that it is a black box is not explicable uh it locks internal causal models um there's not a way of reducing the complexity it's sort of a batch processing there's no real-time updating and that batch processing we seem is extremely expensive and and so we we we're caught on on on that that issue um it's not atic it has no no no means of a spontaneous adaptation and correction on its own um and suddenly doesn't know what it it doesn't know that has no way of representing its completeness of its knowledge and and adapting coherently to that and and and also I would say it's not it's not derived from sort of fundamental scientific principles it's really sort of an engineering phenomena on a case-by casee base uh basis now this is changing um and I think the same issue if you start to see foundational issues for multi-agent uh llms in other words the transition to to making agentic architectures where you can um create a particular agent that's optimized to do a particular task or role a lot of the success of that agent is depending on the quality of the prompts of that and sometimes those prompts are good sometimes the prompts are bad but there's not a really principled way of doing that currently and so they have a very fixed way of of of and specialized kind of goals it's sort of a a a mechanical it's it's so the instantiation of an industrial model um that we see um now what we we rather than having a a a sort of biological self-correcting model it is s of these fixed static models now what I would say the active inference difference here um is is that it's really developed from first principles and I think and and I think you're going to see the Andrew is going to go into this much more detail but it really C it cuts across different disciplines that themselves have been uring fairly quickly in the last two or three decades and so it is going in basic basic physics Quantum information Theory the a whole idea of computational Science Biology and cognitive science all these are intersecting into creating a new way of looking at what intelligence and life are and how they're interconnected um and it also builds in wellestablished statistical methods including Markoff blankets blazi and belief mod the stff the work has been done with JD a pearl um and basically what you at active infine does and and just to highlight is it codifies a predictive brain thesis that's been nurturing in in neuroscience and so the pr and then that in turn sort of mimics the processes of the scientific method and when I get so we think about what the method is that you're you're trying to codify and and I turn to Richard Fan's comment well quoted comment that I can't what I can't create I can't understand so there's a new kind of criteria for what it means to have a um to understand something is be able to model to create a digital twin if I can recreate something then I understand what it is and if I can replicate that behavior and that that's sort of the the the principles that you find in active inference from the very beginning is ah how do I grant a generative model how does a living agent have a generative model of it s in its world and this this really owes it work his Origins back to to uh the work that's done with J Pearl on his understanding in causal reasoning and also markof blankets you how does some what is an autonomous thing and this is where Carl fritzen anal work how how does something maintain its separation from the world in which it is in and so it is not governed by that things but maintains its autonomy and there's a certain kind of topology there that and something has certain kinds of inter internal States it has external States Observer States and action States and there so there's a lot of deep Concepts that's in it

I'm sure Andrew will go into that much more detail um and then what we have is is the active invers agents themselves as sentient live things so you're creating something and there very different than say a large language model you're taking something it has agency and it has a representation of itself in the context in which is trying to stay alive um and so it's self-organizing it's self-healing is self-replicating and symbiotic in the sense that it evolves complexity Mutual synchrony with other living things in its World in its Niche it's a joint Niche construction uh hence sentience is is sort of a is a kind of situational intelligence an awareness that's granted in the survival of that particular agent and the person that's sort of the the context of the person who's really behind this is Carl priston and and um my own experience of being someone who is very interested in sort of and artificial intelligence from from many many years ago and self-organizing systems as cybernetics it really hasn't been until the last five or six 10 years that this has come together in a coherent uh not only coherent Theory but a whole coherent formalism um and through what what car would call physics of living things and he's one of the most SED sists in the world it's a neurosci is has done um basically developed a reputation neuroimaging and infer the structure from the brain from Neuro Imaging uh and he applied the free energy principle which we'll go into more discussion to later um and so one of the I love this idea of a tree foil knot it's sort of like okay is it is sort of this autocatalytic biotic system it's really important it's a living system and it has certain kinds of characteristics that are all interwined with itself one cycle synchronizes is exception or it's a representation external world to minimize uncertainty the other does it internally it has to align that with the external world and then you have another cycle that has expected free energy minimization it makes predictions about how to change itself in its World um and so what we have here is basically a a a what I think active iners offers is a really a soundness of grounding in bi in biology and physics and so you have a sort of a ground truth of evidence space that do I make predictions and do those predictions allow me to act and survive in the world um so it can be explicit it can be explainable and and we'll see we'll talk about this later but you can have since it has cause of models and language and knowledge graphs it can explain why it doing what is doing and also it can talk about explain why how much it doesn't there degrees of uncertainty associated with it and there's also the potential of self- policing self cor acting mechanism and I think this this is really important in other words you can design a active inference agent to to live with in certain constraints and Bounds and you can build in certain me me mechanisms that have for self-governance that cannot be captured uh so it's actions are within the limits of its designed Mission and not to have generate negative externalities so this I will sort of leave with this one and then just to quote William Blake I mean who talks about this a poetic form is a fractal nature reality see the world in a grain of sand and heaven and a wildflower and hold Infinity in the palm of your hand and turny an hour this just sort of pulls it together in a more poetic sense so on this I would hope to uh hand off to uh to uh Andrew if uh thank you I great thanks John um and so hi everyone I'm me Andrew um Andrew pase I'm uh currently involved with the active inference Institute it's been great partnering with with uh Dr clippinger with the director of The Institute you know we've we've been doing some really exciting work lately um and I want to address a lot of the points that that John has made although most of that will come tomorrow when we give some major updates on our biofirm so today in the spirit of like an initial workshop and kind of introduction to active inference I'm actually going to refer back to this tutorial that I gave a couple months ago uh at a International Conference for computational social science meaning folks who were involved in sociology economics uh you know a broad range of fields and uh the context was applying active inference a lot of these principles that John is referring to to social science research uh and and I I basically make the the claim and the argument this is already being done so I'm not necessarily providing a bunch of new information but it's just to kind of extend the dialogue to computational social scientists like here's how we can apply active inference to you know your particular setting in different ways so uh we'll just kind of cover some of of what I went over and you know I apologize if there's any Whiplash from how quickly I go through these slides um but we're just going to move a little quickly um so the cont context again of this this uh this tutorial that I gave was it was split into three parts it was a 3.5 hour Workshop where we actually we started with talking about agent based modeling you know how it's traditionally been done and then followed by that something that I would call a cognitive turn in agent based modeling that's happened over the past couple decades as you know there are these kind of calls for you know we have the computational resources now we have an immense amount more of you know Empirical research and validation of different kinds of principles related to human cognition coming from Neuroscience neurobiology uh you know cognitive modeling uh studies of neuronal Dynamics uh organisms in their environment all those kind of things so so there's a shift to that and then uh the most recent you know means of trying to establish that the broader state-ofthe-art area is uh using reinforcement learning agents so we'll touch on what those are uh and kind of how that work works and you know anyone who's kind of been looking at the the agentic uh framework for for what's being done today they'll probably already be familiar with reinforcement learning then after that we'll go into active inference and how that kind of kind of bridges together a lot of these different things um and apologies I want to just mention real quick uh you can actually go to this this link uh the quickest way of following that is just by going to Google type you know my name Andrew pase and then you could type GitHub or ic2 S2 in any case you'll come up to uh the landing page for this you'll see links to all previous slides I've made uh slides that I'm making for tomorrow uh I'm also making a lot of his Open Access uh so so there's code that people can you know play around with and things like that um so agent-based modeling uh this has been done you know this goes back to roughly 70s 80s with a lot of you know uh initial developments of computer technology and being able to realize that oh we can we can simulate uh people right in some loose sense in in some kind of simplistic sense uh how can we simulate people and why would we want to do that we would want to do that so that we could maybe imagine uh hypothetical scenarios imagine you're designing some kind of policy or want to test a policy uh and see you know oh if I recreate that in a simulation and have people in the environment where I'm implementing that policy uh what happens what's the collective outcome you know what occurs so all kinds of fascinating work from uh shelling and the the traditional way of looking at uh housing segregation for example th handling those kinds of problems where do people land where how do they settle you know in in in uh like a residential environment uh even more interesting work later after that Santa Fe Institute creating an artificial stock market uh where you where you have agents who are basically interacting with this kind of financial environment where they're looking at stock prices changing they make choices to you know buy or sell Etc and see how that plays out Recreation of housing booms and busts excuse me not housing Financial so this can be applied in many different ways point and so what what is agent-based modeling you know in its fundamental principles and how does that relate back to active inference which we'll get to um key components of agent based modeling is that we have agents kind of the catch word of the day um traditionally agents are you know that they just represent some entity a living organism you know this this can be used for for biological and Zoological studies trying to understand you know how ants or or insects you know interacted environment or at a broader scale how humans and entire you know societies and cultures might interact with their environment and and so what's really important about this with agents environments simulations is that know we just kind of run a simulation over a series of time steps what and so we let Dynamics play out and then what is it that that composes those Dynamics it's the assumptions we make about agents what is their behavior what do we think that they do you know how do we program them in this way or in that way uh and then also what are the aspects or characteristics of the environment you know and so again given that a lot of this work was done starting in the 80s going forward with with with very you know we nearly archaic at this point uh forms of like computational resources it's like things were kept as simple as as possible and there are a variety of guidelines that need to be used to develop agent-based models for running these kinds of simulations uh you know and so so some of them even running up to to 2007 here this is the work of of doctors laser and fredman who are respectively uh have been involved with Northeastern University in Stanford um you know that they give some guidelines for how to do this um guideline one and two uh in some is you need realistic uh assumptions that you're making about your agents but not too realistic in the sense that they're too complex one because we don't have computers that are Advanced enough to run highly sophisticated complex agents and then also you know if we make if we make too many assumptions then the situation won't generalize well you might be making an agent based model for one very very particular scenario where it it's not adaptable across different real world circumstances right so we need some kind of balance in terms of the complexity of the assumptions we're making the Simplicity of the assumptions we're making the rest are a bit more plain which are one uh reproducibility you should publish your code you should document it very well other researchers should be able to kind of take up the torch and be able to experiment with your code um you know we at the Active inference Institute we're constantly putting out new get Hub repositories and documenting or processes so that other people can come get involved so I just you know want to mention that there's a big problem in reproducibility in science today and it's very well known you know so uh just wanted to touch on that the nont reality it's like why do agent-based modeling uh if it doesn't help us uncover anything you know why why not just sit down and have a conversation about you know what what do we expect to happen what are our thoughts about what's going to happen why you know do all of this um with with setting up these programs and and agent based models and designing them if we can just come up to the same conclusions through speaking so you you want your results to be interesting you want to make realistic assumptions that you know we can kind of have all these things synthesize um together so I made some points about balancing Simplicity and complexity this is just a representation of uh you know this these These are common problems in machine learning and deep learning you know you you go and get uh you know University education these are the core components of all of your courses you know this isn't specific to active inference like wide ranging you know concern how do you uh create a model that is realistic enough capture uh and and explain the en the phenomena you're looking at um but uh simple enough to where it's adaptable to different circumstances you know there you know these are kind of in part uh you know just as uh Dr clippinger referred to Richard feeman you know I just some reminders from you know famous statistician George GP box you know all mod all models are wrong but some models are useful right we're always trying to approximate a reality and so the question is how do we come CL closer to approximating uh our reality and so how do we actually you know using modern computational resources using all these new advances in all these fields how do we actually make you know new forms of what we'll call cognitive agents who exhibit the kind of mental Dynamics and social dynamics between them that actually better approximate real human behavior so I give some definitions the differences between traditional agent based models and what agent assumptions are are being made and uh more often than not for the sake of Simplicity an agent is pretty one-dimensional if they observe X then they do y otherwise they do see these very simple pre-program roles they're not autonomous they're they're kind of like you can think of them as like thermostats or something like that right the temperature is too high you you turn it down if it's too low you turn it up um that kind of reactive framework uh there's no beliefs about what's going on there's no internal goals per se aside from a single homeostat that is you know maintain the temperature at what you said um cognitive ages however the expectation for those this throughout the the recent uh agent-based modeling literature is like how do we achieve cognitive ages we exhibit internal cognitive mechanisms they exhibit perception action they actually learn and update their models I'll make a point that you know as John referenced with in the case of llms which I'll cover much more deeply tomorrow llms do not I mean it's so expensive to train them and you you typically just train them once right and so um they don't learn in real time they're kind of stuck with the knowledge that they have they have a lot of it to be fair they've been trained on you know thousands thousands thousands of documents uh and so on um but but that's it like that's you have and then and then we have to figure out how to work with LMS from there with cognitive active inference agents as well as other kinds of cognitive agents the ideas that they can learn in real time meaning in the moment and adjust their expectations their beliefs to what's actually going on now um and then also planning so the capacity for being able to infer or guess using evidence that they take in like what's going to happen in the future right and that's immensely important in today's climate right where we're that I mean that's in part the reason we're doing in face model is to attempt to plan to test things to to come up with our own inference about what's going to happen in the future and then act in actionable ways to to realize our own goals in the future right so that's that's you know ideally that's what an agent would do and be capable of um and then with you know this cognitive turn we we're also looking at a broad synthesis of all these fields you know as John mentioned we're looking at things like cybernetics and machine learning we're looking at physics information Theory um people are starting to it's great it's it's fascinating that so many uh interdisciplinary conversations can come up and collaborations between researchers of various backgrounds because it's like oh you know once we once we realize what it is we're trying to do here we can start to pull from maybe principles or relate principles from other fields to this field what works what makes sense what's empirically you know what can be validated so on so it's it's um you know it's a very rich field right now and then furthermore a lot of traditional agent based modeling has been done using uh uh software called net logo which is very specific uh it's it's good if you want to do agent based modeling like you just learned this one programming language there's a GUI like an interface for using it highly simplistic it is not integratable with other tools not integratable with you know llms with natural language all those kinds of things um and it's so it's it's very narrow you know it's good at what it does uh with cognitive agents we would want to be able to pull from all kinds of other resources that are you know available including open source ones uh being able to use apis being able to pull in other kinds of real world data not just purely simulations you know so um there's just a lot of richness behind this idea of moving to cognitive agents then there's some basic you know uh ideas of what makes a cognitive agent again perception they take in observations and they infer what's going on in the world and kind of like we what we do my stomach growls and I infer that I'm hungry for it's a very simplistic example um and then agents are are generative the sense that they actually come up with uh their own plans or a you know choices of what actions to take based upon what they believe such as I observe my stomach is growling I iner I'm hungry I will now choose to go eat food as like an affordance that's available to me in my environment so this is a new way in a certain sense of doing agent based modeling where before we were looking at modeling an environment with a bunch of one-dimensional Agents you know predictable agents on the inside if they see X and they do y otherwise they do Z but now we have models in a model in the sense that we're actually able to internally inspect our ages right by having this beliefs based frame framework where agents observe real data or information and then infer beliefs about it we're able to then retroactively look inside the agent and see what its beliefs are as opposed to these kind of you know very difficult to to interpret uh blackbox models that we find very commonly in deep learning and and and you know very very highly performative yet very uh uh difficult to interpret like neural networks and architectures like that with with uh agents we would want to be able to see in a in almost like a human sense like what are their beliefs that they're holding and why are they doing what they're doing so we need to be able to accomplish that in order to actually realize all the additional benefit of doing this which is being able to look at you know internal cognition what are agents learning whenever I whenever I Implement a new policy or something right I want to be able to know what's going on with them internally I want to know the full impact of of what it is you know this plan I I'm considering like actualizing so let's bring it back um you know how how do we address all these traditional criteria how do we make realistic assumptions how do we um you know how do we make our code reproducible that's an easy one because like well we're still coding the whole time uh and then you know how do we how do we derive useful insights from this um and what are other problems that are going on like now that we're getting into cognitive architectures um and and we we are starting to brush up against the potential of recreating more blackbox models right we need to keep things simple enough like how how do we navigate all this a common problem in in social science research and general but also in data science machine learning and those kinds of fields uh reinforcement learning common issue is what is h how do we handle What's called the explore exploit tradeoff this is this is famously an issue uh if you have't agent who comes up with a suboptimal uh solution to a problem you know let's say an agent who's you know they're trying to figure something out they find something that they deem to be good they get a positive you know reward signal or positive feedback from that but it's suboptimal there there's actually a better solution exists they just don't know about it yet how do you get an agent to actually be willing to explore same thing for humans right we we don't always stick with the you know the first decent looking you know option available to us sometimes we we explore in the sense that we seek better solutions to problems and this is especially important and something to be addressed in agentic Frameworks where if you're trying to use agents for say software application or use case like you want them to be able to find better Solutions ideally you know at any given moment especially in a changing context that might call going forward for better Solutions you know say agents who were maintaining an environment that's subject to uh natural disaster or or or or climate change right like that that might call for a whole new set of of of policies or actions that need to be taken uh you know that that couldn't be relied upon in the past so we need agents who seek new information um unfortunately in reinforcement learning you know that this is still an issue um people are trying to figure out how to kind of manually Define what what defines exploratory behavior and it tends to boil down to kind of simple mathematic of of defining like okay well we'll s we'll we'll modify these agents so that 90% of the time they go with the best solution they have they know about and then 10% of the time they'll just pick something random that way they're at least exploring 10% of the time you know you could put it that way active inference again I'm sorry I'm kind of zooming through a lot of this call this is over three hour tutorial um but uh rein with active inference ages this kind of gets naturally naturally resolved in terms of What's called the free energy principle which we'll be getting into um but basically the idea is if you're able to to have agents who naturally seek information perhaps in line with how we do right that when they come up against something they see oh there's value in seeking new information to find even better Solutions the ones I already have and being able to to do that in a way that isn't some kind of ad hoc rule of you know the 90% versus 10% as I mentioned that's that's um kind of a key component you know a key consideration designing agents um so this brings us to the active active inference which I I describe here as a framework for doing agent-based modeling it uh begins with the free energy principle uh that I I'll make a lot of mentions to this nice textbook that Thomas par Carl friston and you know their other co-authors published a couple of years ago um but you know there perhaps even some more updates to be made because it's such a lively field but uh that textbook really lays out a lot of the core principles and and ways of getting started with active inference um the the free energy principle everything that changes in the brain must minimize free free energy now despite all of this energy disciplinary communication dialogue between you know again cybernetics with psychology with behavioral economics um that actually sounds quite simple right to be able to boil everything down to oh we have a brain it's trying to minimize free energy that that that almost sounds simple enough to where we might be able to employ that as a kind of you know simple assumption that we make an agent based modeling it's quantifiable uh it's it's a presume Pres assumed is the wrong word uh but but something like a principle that can be applied in any given circumstance whenever we're looking at you know the these organisms who are self-organizing such as humans such as you know living organisms such as you know plants in an environment that's being taken care of uh you know that they exhibit this this kind of uh characteristic of minimizing free energy as a as a form of quantifiable uncertainty so we'll touch back on that again in a moment but the the the what active inference provides is a kind of normative holistic framework for what I argue should be applied to agent based modeling it it itself addresses agent based modeling that phrase is used in the textbook this is not a foreign you know out of left field kind of invasion of active inference and agent based model and proposing there's so much overlap and I think it's just an underutilized connection that's already been made right um active inference is based in you know it it provides neural process theories meaning that we're not just you know assuming in the same way with agent based modeling we might assume oh all people are utility maximizers um in this case we're not just assuming it we're we're looking at neuronal Dynamics we're looking at you know Decades of research into using neuroimaging studies and looking at how you know how does the brain really function how does it handle uncertainty different kinds of Behavioral Studies and tasks where you can you know record uh MRI or EEG along the way and look at the Dynamics of those um uh of that data that's being collected along the way and um and then from there we can start to kind of model how that works so um as I already mentioned uh we're still looking at a kind of belief-based framework which is to say uh you know we're looking at uh for for agents who have beliefs and they update their beliefs and they do that by way of the free energy principle which is minimizing uncertainty uh we we start we start to ex we start to see these organisms that we're modeling humans or otherwise we start to have naturally emergent adaptive Behavior right by by trying to figure out what's going on in your environment and then adapting your actions in order to better understand it to better realize your own goals you're effectively minimizing your uncertainty and uh the free energy principle basically is a kind of free energy is like a proxy for that uncertainty that that renders a lot of these things computationally tractable meaning that we're able to quantify it we're able to report it whenever we run agent-based modeling we're able to like actually look on a chart and a graph how much free energy is being reduced uh over time you know and and look at the agent internal beliefs about you know what what is it that they're figuring out uh within their environment over of course is simulation um the view from from this perspective is that uh another aspect I I already addressed the expl exploration exploitation balance and tradeoff uh self- evidencing this means that agents in order to minimize their uncertainty um one way that they can do that is is um to try and act to realize their goals there's something they want something that they expect right and so it's as if you have this prior set of belas of what you want to see in reality this is the you know it's this is the kind of framework that active inference deals with but also referring back to you know Juda Pearl as clippinger mentioned uh you know this is a kind of basian idea you know this is rooted in something like like basy and uh statistics or even mechanics where the idea is that you you have your prior knowledge you bring it to the situation and then that will impact how you update your beliefs based on new information right so we're we're not always starting from some blank State some some starting from scratch we're always coming to a situation with what we thought before and then that gets updated how does that get updated from active inference that standpoint it's through free energy principle um so you know it's I made some points in this tutorial because I realize we're we're we're really uh uh you know I want to make sure we have time to to kind of bring this back um but uh with active inference that we already have so many aspects of uh reinforcement learning and active inference which overlap we're trying to create cognitive models you know active inference I don't want to say it's necessarily competing I don't know if that's the right word but it's you know it's it's it's making advances that reinforcement learning is trying to make itself so um and uh and with active inference it's like well we with Act inference we're actually basing this on you know a variety of of of principles you know that again sorry zooming through you know we're able to like cognitively model you know what's what's going on in the mind in various ways that reinforcement learning is really lacking um today it's it's really lacking that it's it's uh you know these models are being built to be highly performative but they're missing the mark whenever it comes to something like realism and basing uh basing these reinforcement learnings and something like a you know science that extends Beyond just the mathematics of machine learning right we we want to be able to find biological physiological substrates for how all these things happen right um we want to be able from there to be able to apply everyday words and interpretability so what is it that defines what a habit is what what process is involved whenever a goal is formed and made and then from there uh you know attempted to be achieved through action uh you know what defines planning what is prediction um so I just want to see if there's anything else I might be able to touch on this was a nice um you know for for for anyone and maybe I should have started with this but this is a little bit more of an intuitive glance and this is actually these are not words from uh Carl friston surprisingly uh they're from uh Sarah Feldman Barrett who's at Northeastern he uh you know nearly as rep reputable as Carl friston in the Neuroscience field and I just personally uh as someone who's also interested in in in neuroscience and cognition you know I find her work fascinating she's done a lot of work on um emotions and uh you know kind of affective Neuroscience so it has to do a lot with looking at what's going on inside the mind of you know especially humans and so this is kind of trying to provide an intuitive glance that you know how can we think about B in Frameworks about um um inference about active inference as well so it's like uh this is a long quote from a podcast that she did your brain receives sense data from the world sights and sounds smells and so on receives sense data from the body so when there's a loud bang or a tug in your chest your brain receives that information as the outcomes of some set of causes but what caused that loud B what caused that tug in your chest your brain actually doesn't know it only receives the outcomes it has to guess at what the causes are this is called an inverse problem you know what the outcome is you don't know what the cause is so your brain has one uh you know another source of information available to it that is past experience memory um apologies for just like doing a slideshow karaoke here but you can read it for yourself I suppose but the point is and with some of this highlighting it's like we're actually taking account of a variety of things you know we're taking account of interoception meaning observations that arise from within the body or for an active inference agent uh you know it it it it has to kind of model itself you know I mentioned earlier you know if I if I'm hungry this goes beyond the five senses in that kind of perspective right touch taste feel all of that um with this it's like well I need to understand what's going on within my body I need to understand what's going on outside of my body as clippinger you know referred to earlier the idea of a Markov blanket it's kind of like I need to figure out what defines the boundaries between myself and what's around me you know on the one hand I need to have a model of myself in order to maintain it uh otherwise I succumb to the environment around me to use some of the language that's used in uh you know the textbook it sounds a little Bak but you know we're referring to real world circumstances if I you know if I don't take care of myself I will naturally Decay like that is kind of the sort of the principle the world is entropy so to speak right it's like if I don't eat food you know I I I fall apart so I I need to have a model of myself what's going on within I need to have a sense of what's going on without uh and then and then I need to be able to kind of synchronize those things in such a way that I can continue to exist as an entity within this environment around me that's a that's a that's a very intuitive you know perhap perhaps not very well technically described but for now an intuitive sense of of kind of what defines a markup blanket so you know for this is a sort of foreshadowing for tomorrow we'll talk about uh this this really interesting project we're working on for the biofirm uh with with with uh you know Daniel and and and under guidance of of John um you know in that in that sense we're we're trying to develop homeostatic agents uh homeostatic meaning agents who you know their own sense of their survival is predicated on uh the the the homeostasis that their environment around them so we're trying to create agents who take care of and attend to the uh the environment around them as if their own Survival depended upon it you know this is a very interesting kind of uh kind of flip you know where where where you actually want to match and model the environment around you but also take care of it in a way that you can maintain it within those bounds just as you know I have to eat in order to maintain my energy levels you know that the agents need to take care of their environment you know within some kind of range to where it has you know enough caloric energy so to speak um so um I think for now John is there anything that you think I have not addressed that might be useful to to our audience or if anyone else would like me to touch on anything maybe something I skipped or or otherwise i"d be happy to to speak more no I I think you you you you've uh covered some of the really key points that um that are that what I I think is you've done is what differentiates active inference from the more traditional agent-based modeling I think is important because when people think about agent-based modeling they they sort of default to what the traditional models are and I think that you've done an excellent job of sort of uh highlighting what the difference is and what the the the having having that broader perspective that more grounded interdisciplinary perspective you you're able to more not only represent the complexity of it but also provide a a way of modeling agents in within that complexity um and I think that the the the the notion of a the the homeostatic nature of of of an agent trying to live in in conjunction or really harmony with the world around itself is depending upon its survival and its ability to stay alive and differentiate and have complexity but the synony of the environment is really another key point because I uh I think a lot of people they think oh well you may say like navigating a fitness landscape you're just trying to optimize different levels but in fact the landscape itself is shaped you're doing joint Niche construction so it's a much more complex notion than traditionally understood General modeling and I think in in consideration we looking at tomorrow this whole idea of the btic character of model having it based upon principles and biology and nature and having being able to create an economic agent that acts that way I think is is is really quite important um and so maybe some people have some questions about that that that we want to to focus on or not but that I I that's something that that I think it's important to highlight yeah awesome presentation Andrew Ernesto if you have a question also anyone in the live chat is welcome to ask a question no questions thank you very much this is so inspiring well and also just to to put it in context I know this is something that you're very interested in in this whole idea of a btic and economic system and been looking at this yourself so this is really trying to lay down the principles really the the scientific principles and computational principes of doing this and and and uh and we'll get in more to that tomorrow but that that's the the intent here yeah a lot more to say again in our remaining 40 minutes I'm happy to read questions from the live chat I'm also happy to screen share and look at some of the registered participants backgrounds and interests sure okay okay you can unshare Andrew and I'll pull up some word clouds all right okay okay so let it load okay these are word clouds that are generated from the participants registration information and we'll we'll zoom in on them and and uh I'll I'll just kind of mention them and we'll we'll zoom in and uh if anyone has a remark or something that they want to uh like highlight or call out about it just go for it okay so this is a word cloud of the uh 150 or 200 registered presenters overall asking about what is your personal background so the biggest and and and the this is scaled just by number of uses of each term so one person mentioning a term a ton of times might make it large so just to be taken as sort of a representative we see those tiny two letters AI which could mean almost anything at this point and a lot of other terms like computational research systems Neuroscience cognitive philosophy social academic robotics technology practice this kind of speaks to the broad range of topics that that that leads people into joining and at least entering that information on the uh form any thoughts on this or we'll keep on exploring these word clouds okay okay all right again just just raise your hand or or or go for it interest okay well there's I'm interested the notion of community and systems I mean applications the community seems very strong yeah yeah and that that's the the sensor of a of a later one so here we asked what what would be useful and by asking what would be useful and what would be interesting we alluded a little bit to the Dual composition of the expected energy what would be useful what would be utility what would be pragmatic and then what would be interesting what would be epistemic so hearing what would be useful you know here we are in in applied active inference Symposium wondering about how does active inference go from being a unifying framework for ecosystems of shared intelligences and all these different kinds of things that we've heard already and we're going to hear in the Symposium to come so interesting that systems a way to operationalize and really move forward from framings of complex adaptive systems to having the implementations and also the community and the people understanding research models collaborators and then a lot of different topics that are being explored in more mainstream computational science Fields like multi- agent algorithms Andrew alluded to that and kind of brought that up in the context of the the Goldilocks parameterization question the agent based model can be so simple it gives you a great intuition it shows at a a demonstration layer how simple Behavior can lead to complex outputs on the other hand the agent base modeling can go the other extreme where it's so sophisticated then it loses its transferability so that's what people would find useful and then here's what they put for what would they would find interesting applications make sense as that's what this Symposium is all about and again a lot of other ideas thinking about different specific systems research ideas architectures synergies toolkits insights implementations support Collective directions here's asking what would help you in your learning and applying active inference and community looms large other people as well as some of the more tool resource textbook education examples Frameworks interactive tutorials different languages python what does anyone think about this I'm uh a little surprised that we not seen the phrase llm anywhere um I just uh kind of noticed that so one of the one of the points that that I you know have been trying to work out and I hope to speak more on tomorrow is um you know integrating active inference with llms and uh you know there have been attempts to to do this and even even with priston and Par and folks you know that there there's been attempts at trying to pull out an agent's beliefs and then use an llm to process those beliefs like kind of decode them and re-encode them in natural language so to speak it's as if the agent is speaking and saying what it thinks and believes and why it did what it did is you know another layer of interpretability and uh I don't know I'm slightly surprised to not see llms anymore um you know it maybe it's a little too maybe that's a little too on the frontier right now um I know it is in many other areas you know and it's um sorry in the couple minutes we have it just uh you know something that I've been working on personally and this is feeding into the the biof firm is um how how do we how do we use llms in a way that we can actually integrate them with you know all these major priorities that we have uh where where we want to have agents who actually have defined goals right and and and ways of trying to stay alive you know in a way that they interact with their environment in a way that it's as if their own Survival depends upon it we don't want llms which you know again I'm just saying this quickly you know I don't me to sound disparaging but llms in one sense can act as like functionally as Talking Heads where you just send in some text and they try to predict what would be the best text in response they still subject to hallucination to confabulate making things up so to speak even though the answer looks structurally sound you know and and and so the my point is that rather than viewing llms as uh as agents I don't I don't I don't think they are I think that uh you could have an active inference agent who is augmented with the capacity for natural language by having it interact with an LM right so yeah sorry to so deep my own yeah scan scanning across all the registration fields for llm mentions so how does active inference apply to the design of llm systems AI outside of llms better ways of Designing internet skill protocols and governance systems and should converge with tiny ml to challenge llms deep learning so it's it is interesting that that it's uh referenced with only a glancing blow and and uh We've even seen earlier like in the responses to similarities and differences with reinforcement learning with friston earlier about the increased amount of attention and Tool development and the later and more professional maturity of training environments and this is like one of the key fosi for the open source active inference ecos system and for the Institute is to develop those kinds of methods and trainings and all on boardings all these different aspects of of increasing the professionalization so let's look at the challenges so this was what what what are the challenges facing your learning and applying active Influence People interesting the problem problem and the solution models the m math software maths again mathematical understanding difficult physics scaling simple examples you know it's it's it's interesting um because in the larger conversation about what AI is I mean it's it's that that conversation is so dominated by uh large language models and quote generative models within large language models um and and what least I've been observing I tried to refer to my my initial slides is that they're starting to appreciate the limitations of of the large language models moved towards more agentic models and more towards causal models and hence creating aent architectures that do work with large language models but they're still there's not they're still not based upon biological principles that we're discussing active inference there more of the mechanical models being applied over and I I think to me from from a point of view of sort of the um the sociology of science soci it there there seems to be these different worlds that that I mean the number of people that i' I've known sort of in the open AI well they're they're very unaware of active Ence and some of the critiques that are if you if you listen to the Gary Marcus he's got a very pointed critique of of um of of large language models and not being able to achieve abstractions but that doesn't extend to appreciation appreciating what the the opportunities are an active inference and so there are there s of these islands of understanding that they still exists um and I think that's one of the challenges and when you talk about developing an infrastructure I me you need there's huge financial resources are going into the the the large language models infrastructure and where I think that the totally bias on this I think all the the opportunity is totally within the active INF of free energy principle and I think that's where the science is going and yet it's it's still it's sort of in in this isolate World um and I think the only way to get out of that is start to to really start to showing the applications and showing the with results and and uh because uh but it it's it's it's interesting I me to see they're they're different they're different worlds um um but I I think that what's the there's a peing and so the hype curve around uh uh the large language models and and it's going to they're going to descend my bet is they're going to descend into a valley of doubt and you suddenly you're going to see that showing up and the in the Nvidia stock going down and then people are saying you can't you know destroy all the natural resources on run these models you don't have to you know the other and Carl alluded to this in his earlier comments today you know I mean he was talking about climate change and models compression I and I think that's going to be a driver of this um so but that that's some my my observations on on these yeah these clouds is yeah I'll bring in a question from the live chat also welcome Ma just thought I'd bring you on if you wanted to participate in this panel or just chill so Stephen selette wrote is the lack of llms mentions because large language models are different to Dynamic agent belief modeling and you want to take that you could you could you repeat that uh yeah question is the lack of llms in the word clouds here but just more more generally thinking about similarities and differences between active inference generative models and llms is is that because large language models are different than Dynamic agent or belief modeling I I don't want to approach this this question in the in the wrong way but I I will say that that that as far as you know whenever it comes to llms and and dealing with you know they internal parameters are obviously highly quantitative and then they're dealing with with natural language as to and then switch back to looking at active inference agents which you know are highly quantitative in nature and uh typically you know if you're making continuous time models uh you're looking at again quantitative information and then even with discrete models you're looking at you know these kind of discrete values that you could potentially move in the direction of trying to you know make it I think this has already been done uh but uh you know there there are agents who um you know have been used to try and understand like oh we can read individual tokens then move up to another layer and then we have entire sentences and then move up to another layer we have entire you know paragraphs or ideas that kind of thing is done but I think this kind of like llm with quantitative uh grounded agents like that connection uh is still that in my view that's kind of the frontier of where things are at and so it's difficult for people to conceive like how can we how can we Translate encode decode re-encode natural language in a way that's readable by these quantitatively oriented agents and then vice versa how do we transition quantitative beliefs into natural language that's the very thing that I've been uh you know looking at thinking about uh recently I made like a active inference like PDP agent who uh you know they what they do is that they use their beliefs about which action to take and then that action that they carry out is actually corresponding to a prompt to an llm and you can actually take those beliefs the quantitative beliefs find some way of meaningfully interpreting them and then putting that into these blank spaces in the prompt is as if you can have the agent say in real time I currently believe this with probability this give me you know a solution to this problem you know based on so so being able to find this kind of newer versus the quantitative and the qualitative you know it's it's historical problem when working with with data that go back centuries at this point so um it's it's really exciting though it's really exciting to think of how to bridge that kind of Gap and I I still haven't seen anyone you know nail that down this this is the frontier this is where we're at and I think that's kind of the next thing for everyone to really be considering thinking about working on um and I'll give my view on how that could be done tomorrow well um yeah I mean effectively llms don't really encode explicit beliefs or really maintain a persistent model of uh the world um even when we sort of give them a sort of loop by which we um communicate with them that Loop in itself doesn't actually factor into the predictions that is going to do um so they they just lack a framework for recursively updating beliefs which is probably why it's not being thought of here um but also mainly that you know llm agents are only agents by virtue of sometimes being connected to stuff and having um explicitly stated goal but I don't think that actually qualifies them to the agents you know yeah I I totally concur with that I from my perspective I I think that the reason that LMS are are are not looking at that are are not looking at or that whole mindset is not looking at at belief structures in in subjective experience is I think it's a product of a whole sort of objectivist reductionist methodology that um is has been tuned not to look at agentic systems I mean in my generation if if you if you were in biology you and and you talked about agentic systems you you were you were outed I mean that was that that that was verboten and I think the the there's a whole school of thought there's a whole set of of in aonian sense a normal science built around uh a sort of objectivist reductionist non- agentic notion and and and uh and so there it's a new generation that's coming in in both in biology and in and modeling and cognition there's recognizing this notion of an agentic perspective of beliefs and updates and and that's a huge shift I think in in sort of the epistemological perspective of of a lot of what has been traditional science that's really interesting there's also just operationally the difference between taking a very large fixed number of parameters in a given model 70 or 7 or 405 training on existing data and then finding patterns or igen modes or activation um summaries that have secondary interpretability but there's not really an effort of trying to find interpretability at the level of activation of a given neuron it's about looking at patterns and trying to find Collective level interpret abilities like in in the AI interpretability field contrast that with designing from the ground up an example agents where the beliefs are explicitly encoded numerically and so that's a different mode of building and also one that's hinted at in in this 2023 paper with Ma and others that highlights element of explicit semantic interpretability and also explicit semantic interpretability of uncertainty estimates whereas a string of tokens emitted by an llm that says I'm not sure it's sure about saying that but that doesn't mean that it has a given certainty about the reference so understanding what the communication semantics of llms are is incredible and what they do and I think when we look at this Cloud for people's responses how can active inference make a difference in 2025 where there there is llms here many many kinds of systems and that's sort of like a microcosm of what the Symposium is different people with different systems of Interest asking H how do these different attributes of the model that we're talking about the first principles grounding semantic interpretability surprise and uncertainty and its bounding as a central component as opposed to reward how do all those things really play out what does that demonstration look like in in different systems h here's go for Andrew no I was just going to say on that that that previous one you were just looking at I I I really appreciate that kind of a broad-mindedness that that people seem to be exercising I like seeing terms like uh symbology scientist governance um industry I mean those are some very very excited to see people who who want to you know so much work has been put into kind of the theoretical and then sort of like in in in vitro and and in lab and and in simulation kind of work that's been put into active inference so I'm just really excited to see you know with the the um International the the conference the the back of inference conference that was given recently and being able to see like a drone that's flying based upon the free energy principle and things like that I'm just really excited to see more kind of real world applications right and and kind of uh creating uh you know systems and and applications for like real world use cases it's it's really cool to see some of those terms there yeah I think that's going to be really critical so see yeah here's the cloud for how are you applying active inference again systems I I think just reflecting people talking about variety of different systems types but and also reminding me of my own journey to active inference being excited about complex adaptive systems however not really having a software toolkit or a formal way to proceed past okay we think about nested multi-agent systems and then where's the software toolkit and then we're right back to where Andrew showed us on the slide it's like well we can do a really simple one in net logo or we can engage on this Enterprise scale engineering research program and yeah here's a more incremental interoperable way to have portfolios of motifs and generative model components and be assembling and scanning across these different kinds of compositions like three ant nestmates in the desert three ANTM nestmates in the forest and these different kinds of environments being able to compose them and over the last several years seeing a lot of the advances in the category theory in the first principles physics surrounding all these sort of research things and it's like that's kind of the water drawing out and then that's the question about how it looks when it is applied and that's what I guess we convene around in this Symposium and and are seeing differently in 2024 than in previous years that's curious so what what's the change that you've seen in pre previous years versus this year and then so looking forward more of an application Andrew or ma what what what do you think I was yeah but absolutely that that that is I oh yeah please yeah I mean I feel like um the work of Maxwell ramstead C Casper HP I mean obviously Carl but um a lot of people who perhaps came from more Dynamics uh decided to take a real interest in what we could do with something that was so scale-free and I find that it's been really really rich in terms of the different kind of theories we've tried to sort of reforme I know that's what most of my work is is a Reformation of existing social sciences um and to your point Daniel the the problem we still have is that we're missing like these key little plugs um that yes you could couch as you know just different schemes of message passing but you still need some interesting tools to to have a a very good description of the vast vast variety of types of structures that we have Which social sciences have done a really great job at if not intuiting at least describing properly but we still don't have the best tools to make accurate predictions and that's part of the reason why most of our polls when we make them are are n they're useless and that should be the main thing we apply them to is just literally just understanding where we're at not even making predictions just understanding where we're at and we still fail at that so I'm very I'm very encouraged by by seeing this and I hope that we will start valuing more and more interdisciplinary perspectives that are sometimes more technical but also sometimes uh you know more philosophical or um or social awesome also while we're in this in in the coming workshops tomorrow will be more of a Biore Regional biofirm focus and then day three will be entirely with at all on this topic and and I want to show one really operational open science Crossover with llms and active inference which is and I'll put it in the live chat as well here I used perplexity which does an internet search summarizes and translates information using llms so here we'll pick on Alexi there's a a profile a learning plan and a set of small medium and large project proposals about active inference so this is just summarizing expertise and experience here we have a learning plan it's just a starting position but this brings a new low bar to the kinds of resources that everyone can take on in their learning and use as like a syllabus or a curriculum and even as an interactive to you know it will not get bored with you asking to explain Concepts again and again or using different metaphors or making it applicable making it like this poetry format putting it in this narrative context and so even asking how do I start with this or can you help me understand what I don't know that puts such a centrality and and highlights metacognition in learning and then the project proposals each of them structured according to essentially what a grant officer would be looking for and what a what a project manager would be thinking about in terms of small medium and large projects so this unprecedented High throughput way to fuse even scanty input from a participant in a inclusive open science setting into real projects that if enough people think hey that looks kind of interesting then we're well down the road of actually making it happen using different kinds of of funding which is something Michael lennin mentioned a little bit earlier so it's sort of like we can get really detailed and Technical and Architectural thinking and and and finessing the differences between reinforcement learning deep learning active inference and we saw slides to that like when Carl showed the action perception Loop similarities and differences okay active inference doesn't have a reward function it's not optimizing based upon this trained auxillary reward concept it's just doing something direct with likelihoods like that is critical that is what makes the active inference generative models what they are but then looking at something like this which wasn't so accessible one to several years ago that just feels like opening up and unleashing what people can do and about using these as Tools in our Niche and I think it's a total open and unknown what happens when you can get to level of project and learning plan for for an arbitrarily complex field with the patience of an AI That's just going to accept questions all day long like that's almost the bigger applications question again just zooming out from how the architectures and the parameters are similar and different to how are we going to do what is authentic and right and best given the tools we have there's a bigger picture Beyond just holding the magnifying glass to different computational artifacts that's well said Daniel I really agree it's transformative if anyone has questions in the chat or here or any other things they want to mention yeah what's going to happen in the next two sessions John or what would be your goals or preferences for for this Arc of the Symposium for you yeah so I I I think is as Andrew alluded um this the session tomorrow is going to be really on the application of of uh of active agents uh to a a design of a economic agent new kind of firm biofirm that's homeostatic um and the objective of that is to develop a new kind of incentive structures a point to Collective action um that are based upon mimetic principles that I think we'll see address a lot of the issues that people are having in the whole idea of bi

Regional Finance and how to deal with climate change and how to create those incentives so there one of the things I'll kick off with I've been talking to a number of people in the whole fields of of um um regenerative Finance bi where they go bofi um this it ties into the work that was done in Elanor ostrom's work on uh how common pole resource how do you manage a common pole resource how do you incentivize people different scales I really think that and what you identify so the requirements that people are looking for it in order to address these problems the scale and I think what we're be talking about is through the implementation of active inference uh principles into an economic agent that it provides I think a a a reasonable way of of addressing these issues um and that um um and and so that's part of the conversation that we want to have and and and bring together these two these two worlds um then the third session is to really look at this um from a a broader point of view is saying where is this going and this are the people that we have in the panels are very off very much involved with Enterprises um applying it to nonprofits to applying it to S problems looking at what they can do with llms currently and what the certain current agentic architectures are and what happens when when you s what happens when you go to full autonomy what does it mean to have autonomy um in in a in a in a biological sense something keeping alive and what does it mean to build systems that have autonomy this issue of autonomy is sort of fundamental uh when when when people think of of autonomous organiz autonomous cars or decentralized autonomous organizations or the whole idea of U the Central Bank the the the the the principles behind something like a Bitcoin is is like oh that that that is autonomous but is it really autonomous what is this new notion of autonomy if we base it on biological principles um and I I think that's that opens the door to a very different set of discussions than than we're currently having and it also opens the door to new Notions of governance and how we can build governance in to maintain the homeotic states um and Andrew is alluding to that so I mean that that that that that's where I become a an optimist and I'm not an optimist around the extension of llms and and open Ai and the and the tech the tech Giants but I am an optimist in being able to apply these principles to to some of the fundamental issues we having around climate and social equity and economic equity awesome okay in the closing minutes if anyone has lost comments and David's question too if you want to address it Andrew or anyone else any thoughts about how we start to develop a multiscale approach to ethics um I want allow time for other people so I'll briefly say uh so so tomorrow um you know we'll we'll we'll present uh various aspects of the biofirm there will be a little bit of a Redux to some of the things I went through today just to kind of thread The Narrative um and I'll kind of talk about you know homoeconomicus and all these kinds of uh assumptions about human behavior and things like that but uh um I mean multiscale governance I what we're trying to do with our bio firm is to uh you know and clippinger refer to um uh Elanor ostrom's work on the commons and so I I'm going to address that a little bit too but the idea is to be able to have a kind of like adaptable system of multiple agents uh you know and so we have a kind of small Society so to speak and uh you know they're kind of governing the commons around them and by being able to have something like the capacity for one shared goals uh and two adaptive learning such that they can actually adapt what they do and what they believe and actions they carry out have those be localized to the current context which is an essential part of ost's like work is like it's not so simple to just give a singular top down instruction that applies in all places I mean to be able to have that kind of adaptivity and awareness of the local um you know while at the same time having agents who are kind of synced up in these various ways I think that's a one way of trying to approach governance and then from there you might be able to have kind of layered you know you could have individual hubs you know and then you could have some kind of like interconnecting system between them that that extends that to an even broader scale of multiple hubs and interaction between them so just some just some breadcrumbs there um thanks yeah Ma and then John with the last word on this session yeah so I mean the question of Ethics is a bit it's obviously complicated but the the the key part about whether we coordinate around multiscale goal sharing Etc share potention that's intrinsic to poweractive inference right it just is the question that isn't going to be solved is what is good uh because you can always say well you know what is good is good by virtue of being contextually sort of stable but there may be other ways to be stable or that might prioritize other kind of individuals or entities Etc so I don't think we're going to solve that question however what we can solve is the is the global versus local coherence of certain kinds of beliefs relative to what might constitute goodness at given scales and find where the friction points might arise because this version of goodness is incompatible with this version of goodness and therefore Uh something's going to give somewhere an error is going to have to propagate so I think that's going to be answered but the the larger broader question you you'd have to look into different schools of thought around what constitutes uh good because we don't necessarily want to go towards um moral realism um because there's a lot of things that are baked into that uh but you might want to go into the uh ethical implications of neom materialism and how it it is about interactions and locality and at that point it becomes about how do certain pockets of ethical framework sort of form and allow to just coexist away from other pockets and what happens when they start interacting uh which again we may be seeing right now in the US yes good point John any last comments on the session oh this is great um and I think U we we now surfacing some of the issues that are really important to to talk about and and um I think Le least well we're going to be presenting is is a is a framework um that at least we can have these discussions in a more rigorous way and in testable way um and uh I think that's part of the way of moving it forward so yeah awesome thank you okay we'll take a one minute break see you soon okay thank you e e